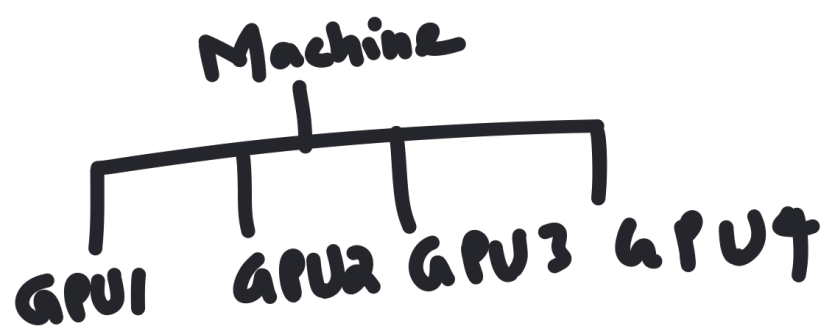




Advanced Finetuning

32 Bit
4 Bytes

VRAM REQUIREMENTS!

Llama 7B $\rightarrow$ 7 Billion

= MODEL PARAMETERS (E.G. 7B) $\rightarrow$ 7 Billion $\times$ 4 Bytes

+ GRADIENTS $\rightarrow$ 7 Billion $\times$ 4 Bytes

+ OPTIMIZER STATES $\rightarrow$ 7 Billion $\times$ 8 Bytes

+ ACTIVATIONS

T

What is in the GPU memory?

- Parameters
- Gradients
- Optimizer State

```
CLASS torch.optim.Adam(params, lr=0.001, betas=(0.9, 0.999), eps=1e-08, weight_decay=0,  
    amsgrad=False, *, foreach=None, maximize=False, capturable=False, differentiable=False,  
    fused=None) [SOURCE]
```

Implements Adam algorithm.

input : γ (lr), β_1, β_2 (betas), θ_0 (params), $f(\theta)$ (objective)

λ (weight decay), *amsgrad*, *maximize*

initialize : $m_0 \leftarrow 0$ (first moment), $v_0 \leftarrow 0$ (second moment), $\widehat{v}_0^{max} \leftarrow 0$

Adam Recap ~~Adam~~ / SGD

momentum $\rightarrow$ ^{4 bytes} $m_t = \beta_1 * m_{t-1} + (1 - \beta_1) * \nabla w_t$ $\leftarrow$ ^{4 bytes} Gradient

variance $\rightarrow$ ^{4 bytes} $v_t = \beta_2 * v_{t-1} + (1 - \beta_2) * (\nabla w_t)^2$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

weights $\rightarrow$ ^{4 bytes} $w_{t+1} = w_t - \frac{\eta}{\sqrt{\hat{v}_t + \epsilon}} * \hat{m}_t$

What is in the GPU memory?

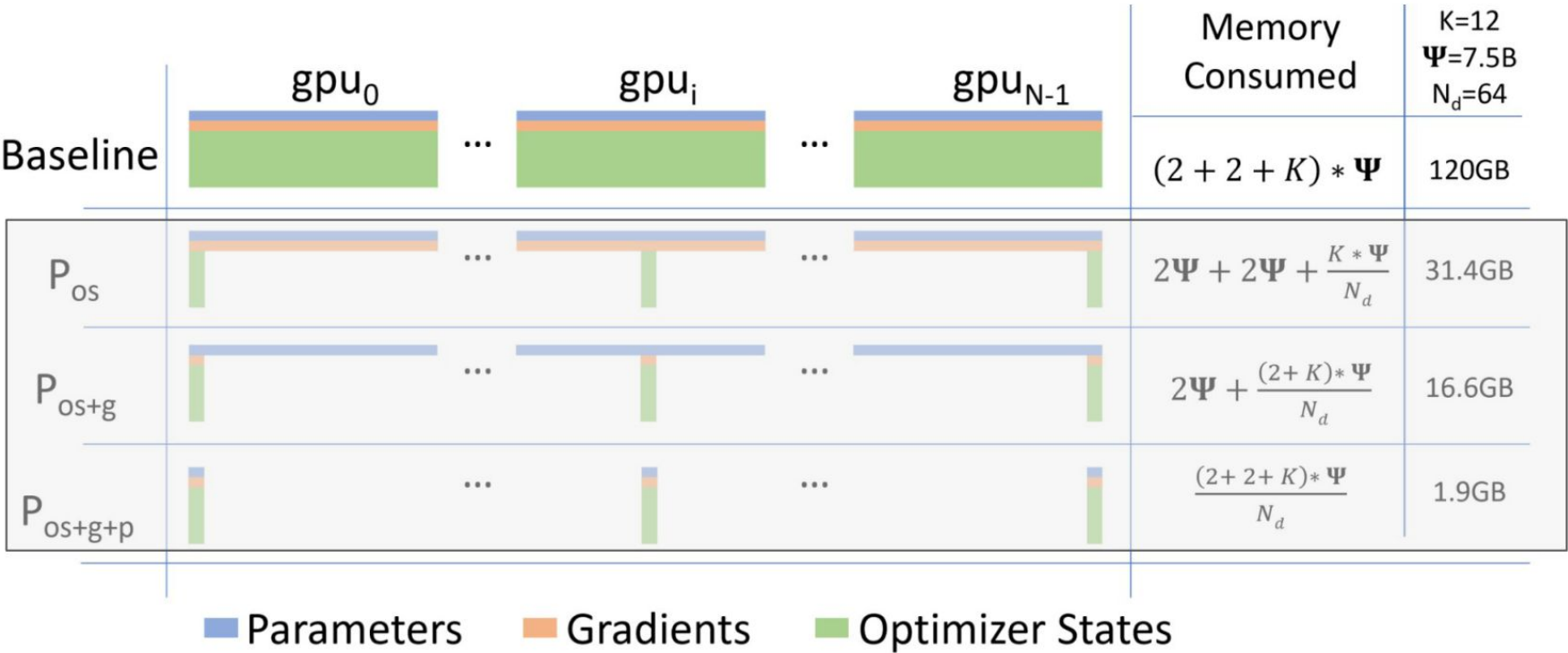
Assuming FP16 and x parameters (*all* trainable)

- Parameters $\Rightarrow 2x$ (2 bytes for float16) $\leftarrow 4B \ W_t$
- Gradients $\Rightarrow 2x$ $\leftarrow 4B$
- Optimizer State [Adam] $\Rightarrow 12x$
 - Parameter copy $4x$ (4 bytes for float32) $\leftarrow 4B \ W_{t+1}$
 - Momentum $4x$ $\leftarrow \left. \begin{matrix} 4B \\ + \\ 4B \end{matrix} \right\} \text{Adam}$
 - Variance $4x$ $\leftarrow \left. \begin{matrix} 4B \\ + \\ 4B \end{matrix} \right\} \text{Adam}$
 - All stored in float32!!

4B
float32

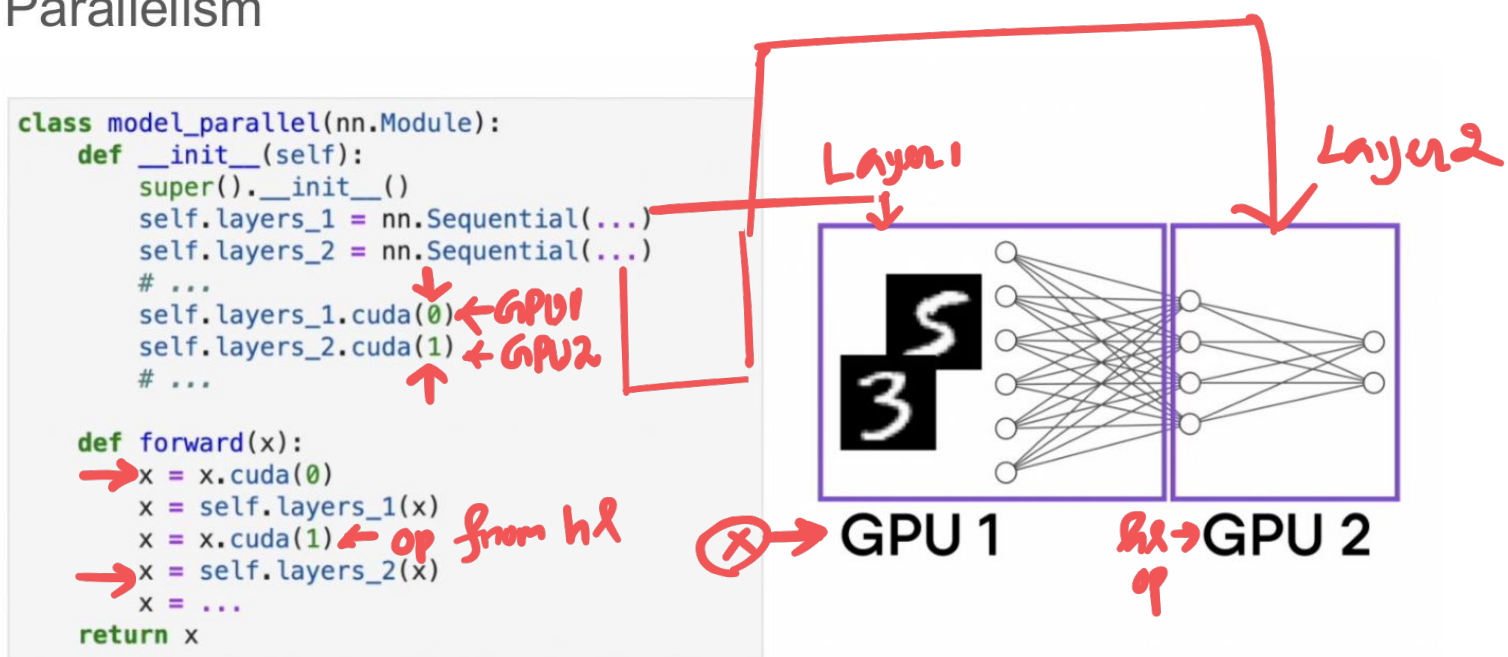
$$W_{t+1} = W_t - \boxed{}$$

What is in the GPU memory?



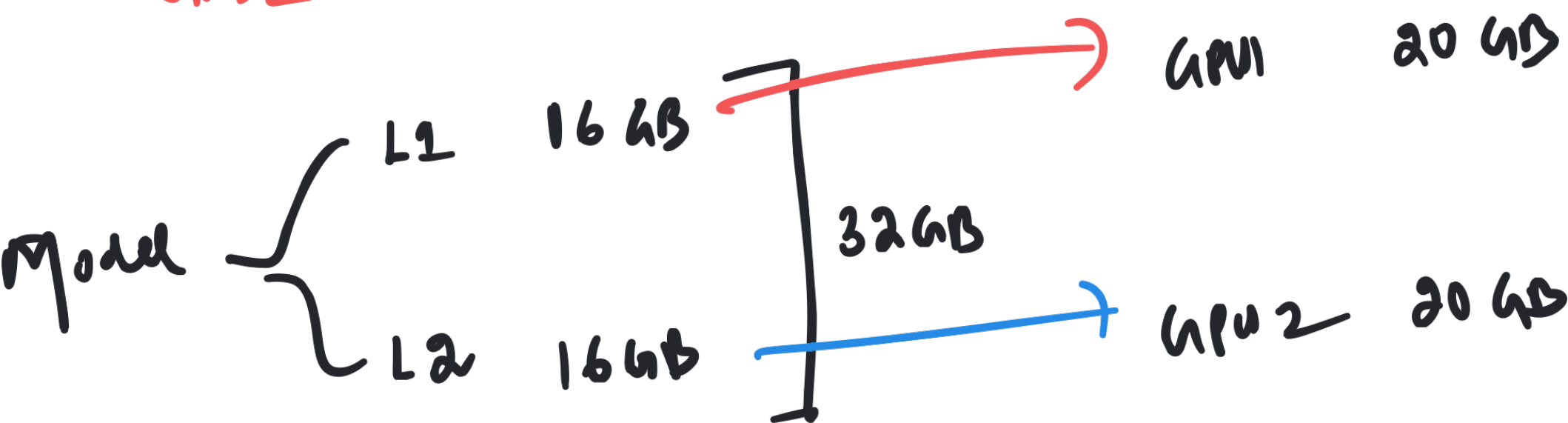
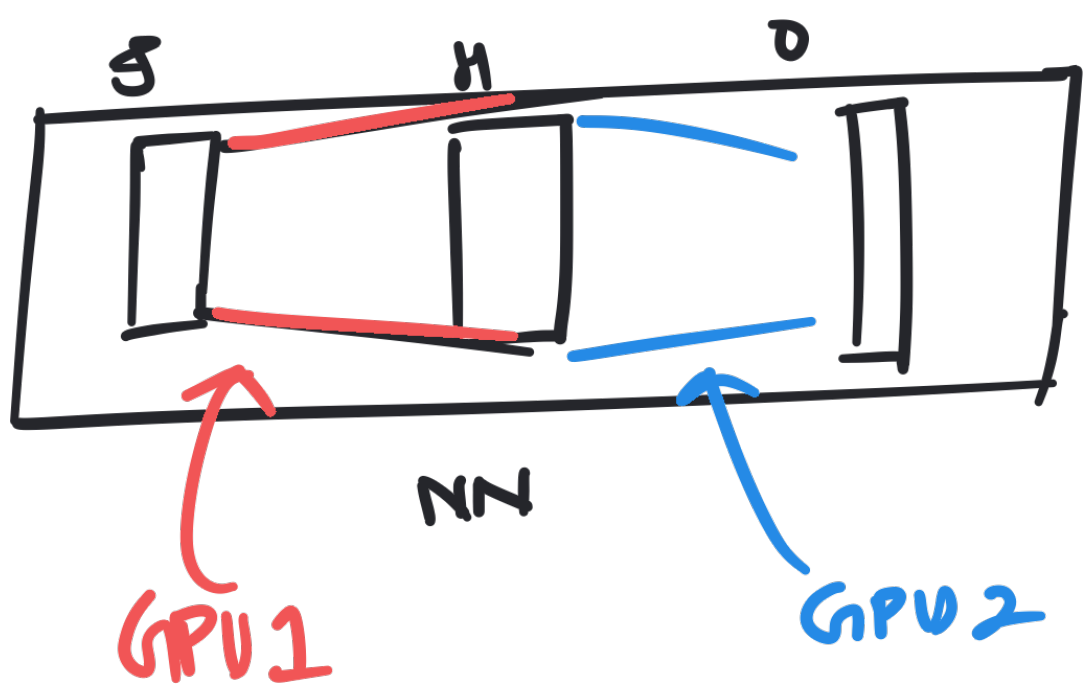
What is *NOT* FSDP? Adv(MP) → wts are shared across GPUs to avoid Out of Memory Error

- Model Parallelism



in PyTorch (not recommended)

<https://lightning.ai/courses/deep-learning-fundamentals/9.0-overview-techniques-for-speeding-up-model-training/unit-9.2-multi-gpu-training-strategies/>



Disadv - Model Parallelism

1] GPU₂ (layer 2) needs
to it will keep waiting

o/p from GPU₁ (layer 1)
(still sequential)

2] Difficult to implement

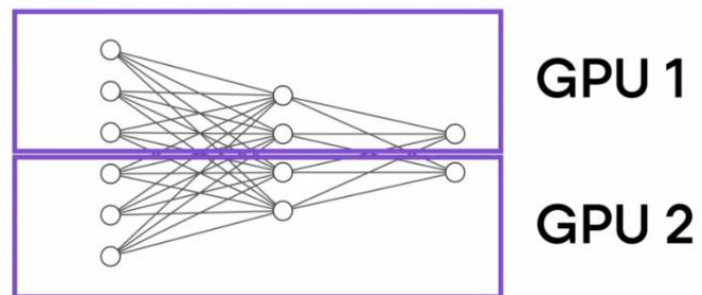
3] Very Poor GPU util

4] Communicatⁿ overhead

What is *NOT* FSDP?

- Tensor Parallelism ← inference

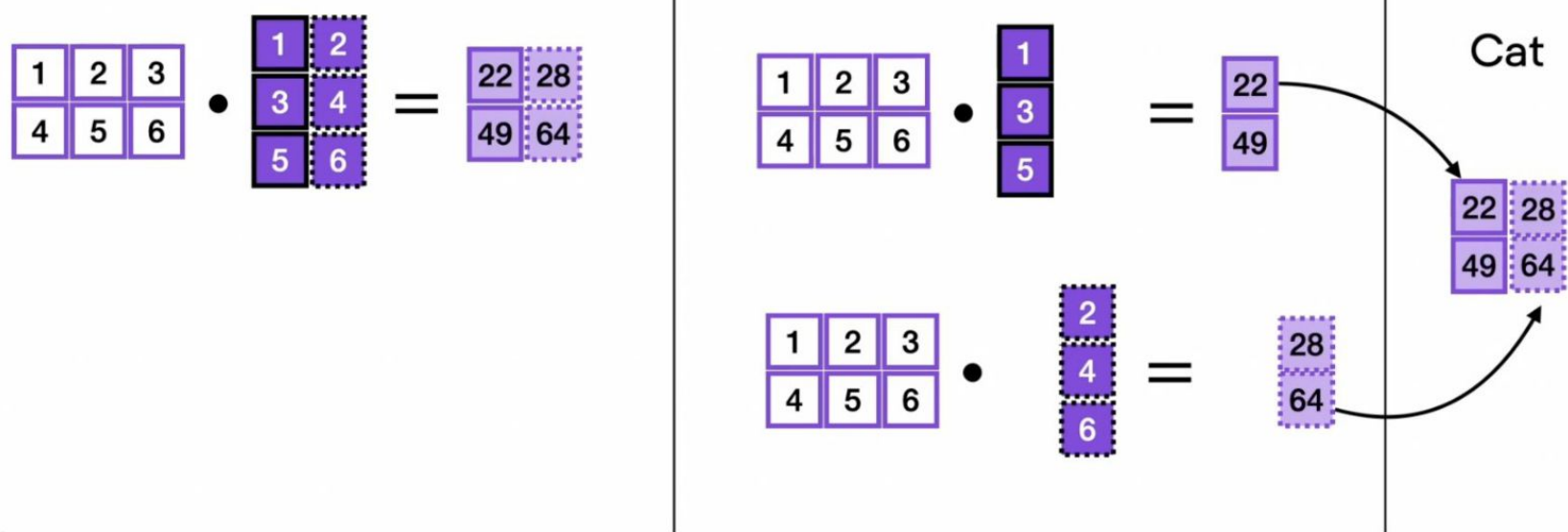
Related to model parallelism, but split horizontally instead of vertically



<https://lightning.ai/courses/deep-learning-fundamentals/9.0-overview-techniques-for-speeding-up-model-training/unit-9.2-multi-gpu-training-strategies/>

What is *NOT* FSDP?

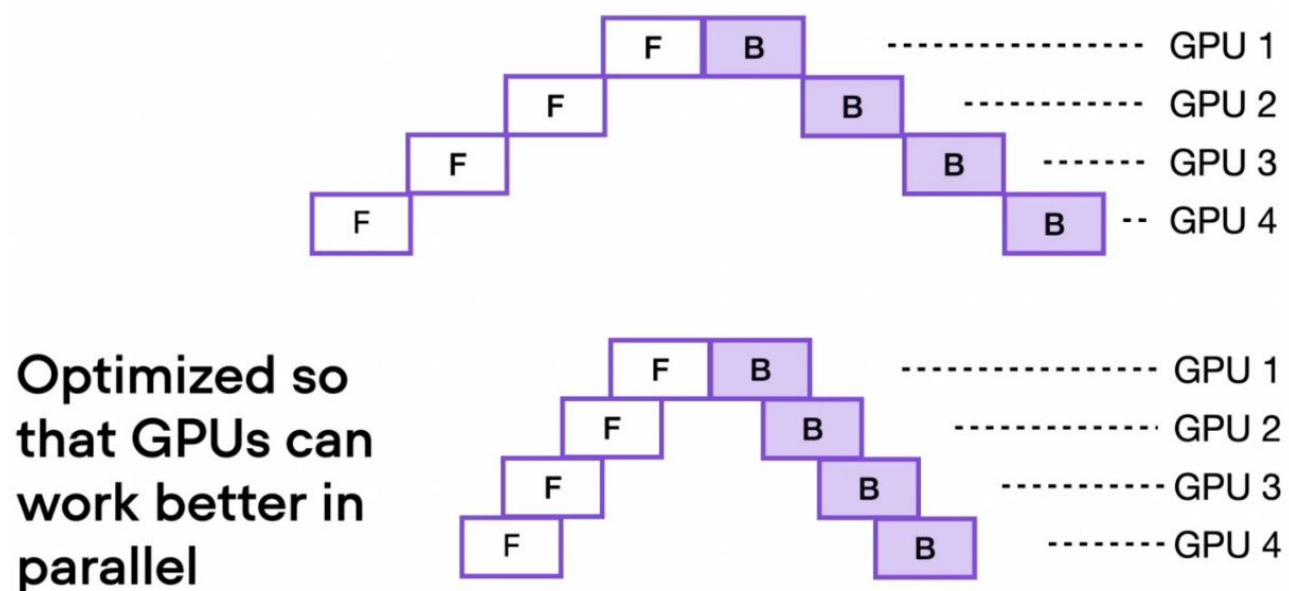
For example, split the matrix multiplication **by column**



<https://lightning.ai/courses/deep-learning-fundamentals/9.0-overview-techniques-for-speeding-up-model-training/unit-9.2-multi-gpu-training-strategies/>

What is *NOT* FSDP?

- Pipeline Parallelism [Mix Data and Model Parallelism]



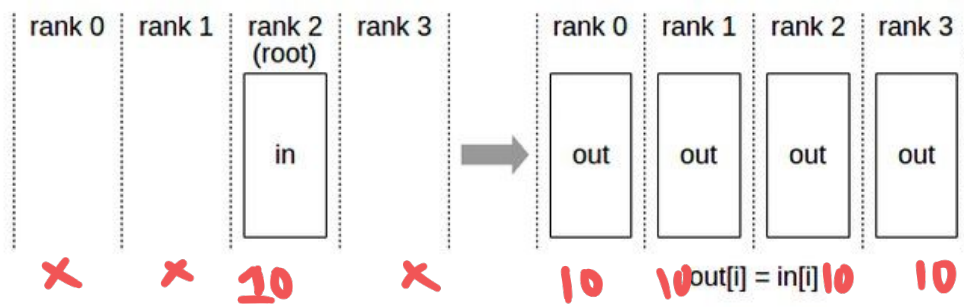
<https://lightning.ai/courses/deep-learning-fundamentals/9.0-overview-techniques-for-speeding-up-model-training/unit-9.2-multi-gpu-training-strategies/>

NCCL

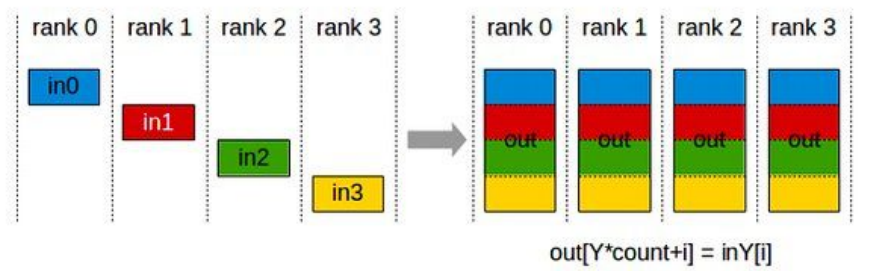
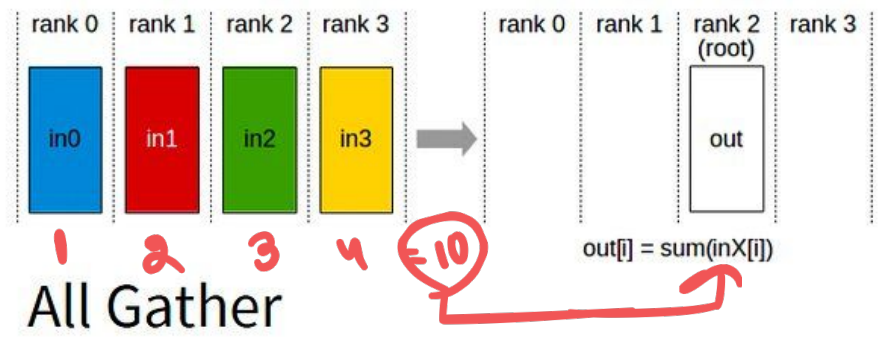
All reduce ↙



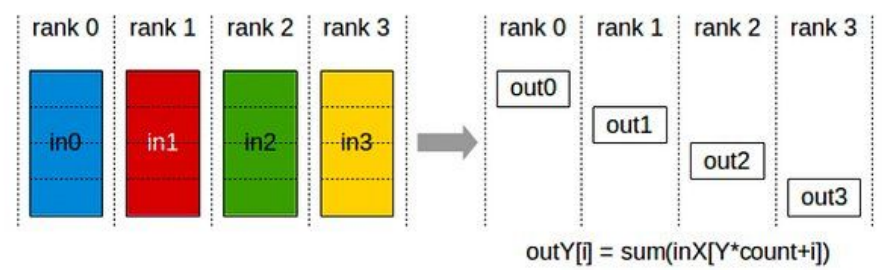
Broadcast ✓



Reduce ↗



Reduce Scatter



All reduce

(avg) gradients

GPU 1 GPU 2 GPU 3 GPU 4

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

~~10~~ ~~1~~ ~~10~~ ~~2~~ ~~10~~ ~~3~~ ~~10~~ ~~4~~

$1 + 2 + 3 + 4 = 10$

Operation - Sum (All Reduce)

Reduce



GPU 1

GPU 2

GPU 3

GPU 4

~~1~~

~~2~~

~~3~~

~~4~~

10

+ 2

+

3

+ 4

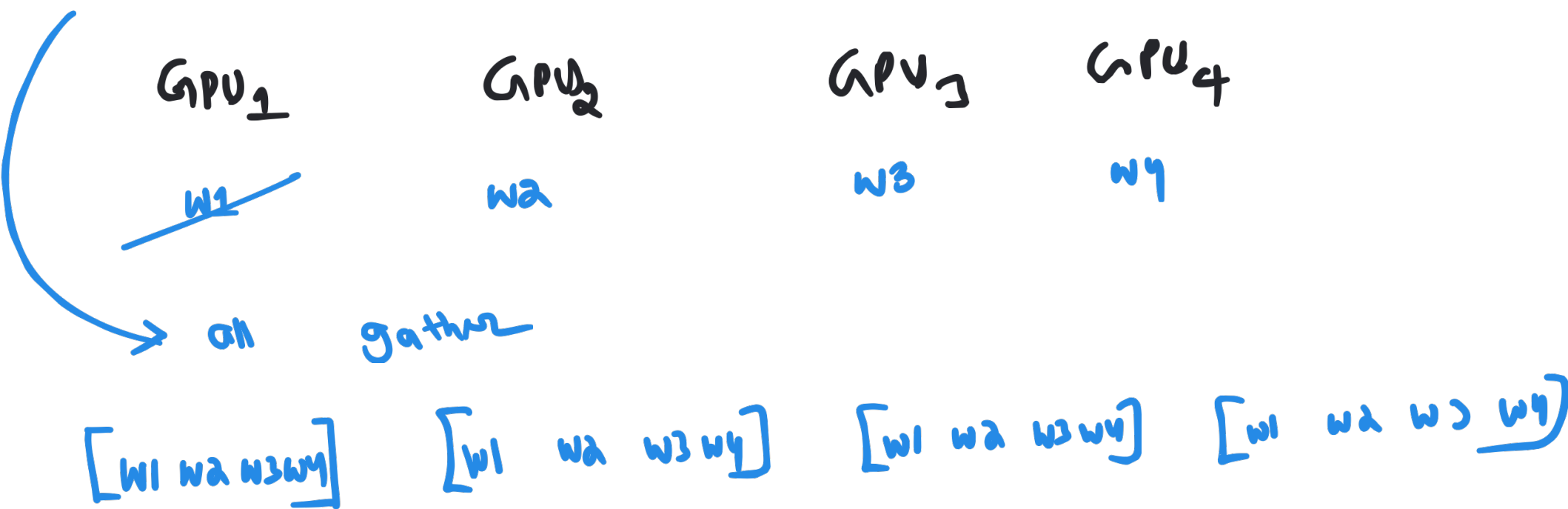
= 10 - Sum

nothing

nothing

nothing

All Gather



All Reduce



Redna Scatter

GPU 1

GPU 2

GPU 3

GPU 4

$[1, 1, 1, 1]$

$[2, 2, 2, 2]$

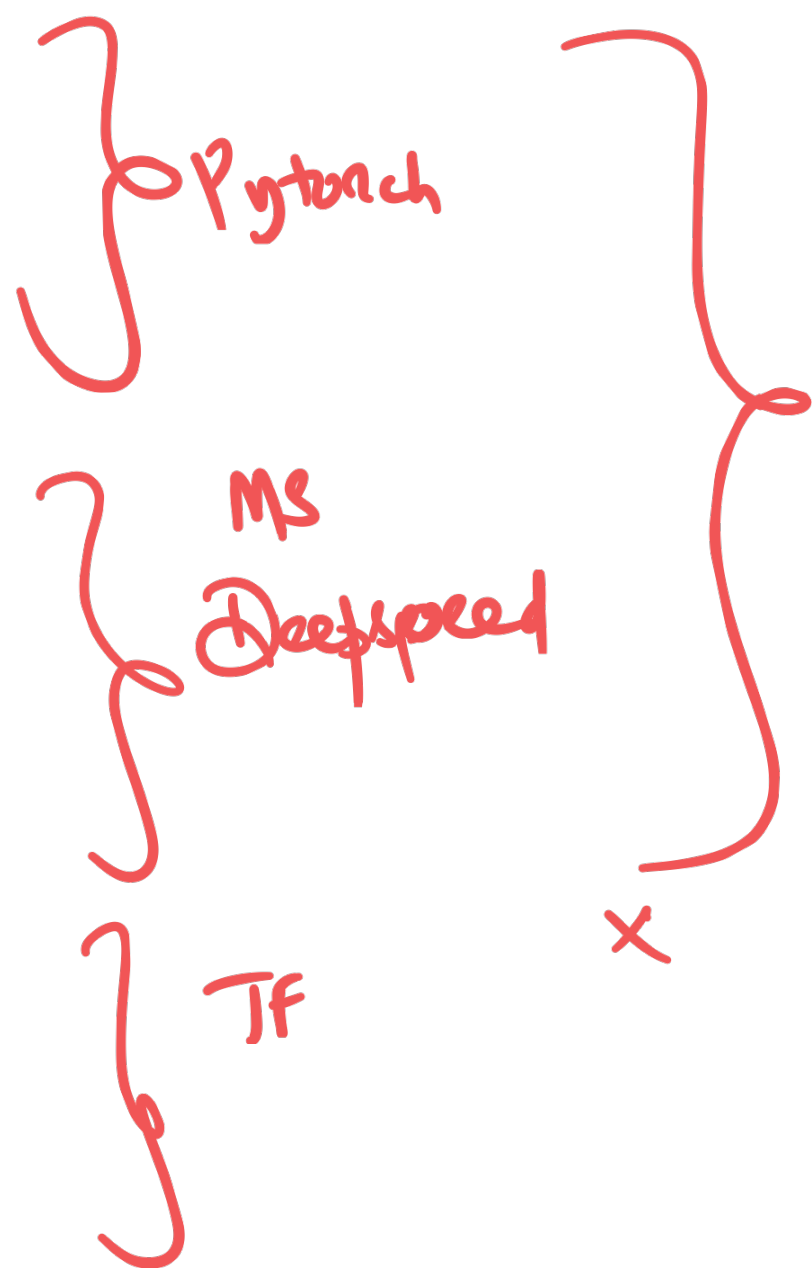
$[3, 3, 3, 3]$

$[4, 4, 4, 4]$

elementwise
reduction

$[10, 10, 10, 10]$

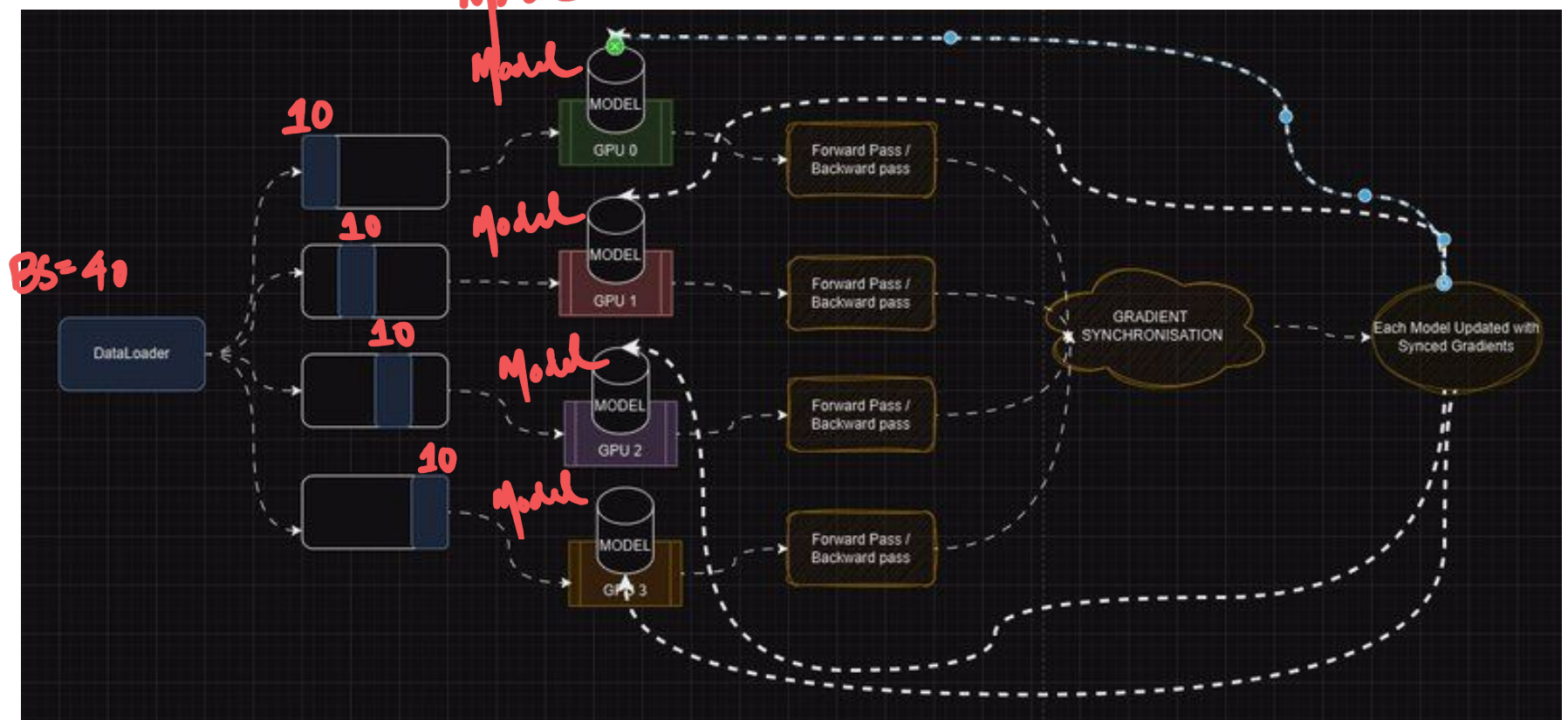
- 1] Distributed Data Parallel
- 2] Fully Sharded Data Parallel



9:20

Llama 70B (Model)

Model cloned



Model + BS (40) → GPU → OOM

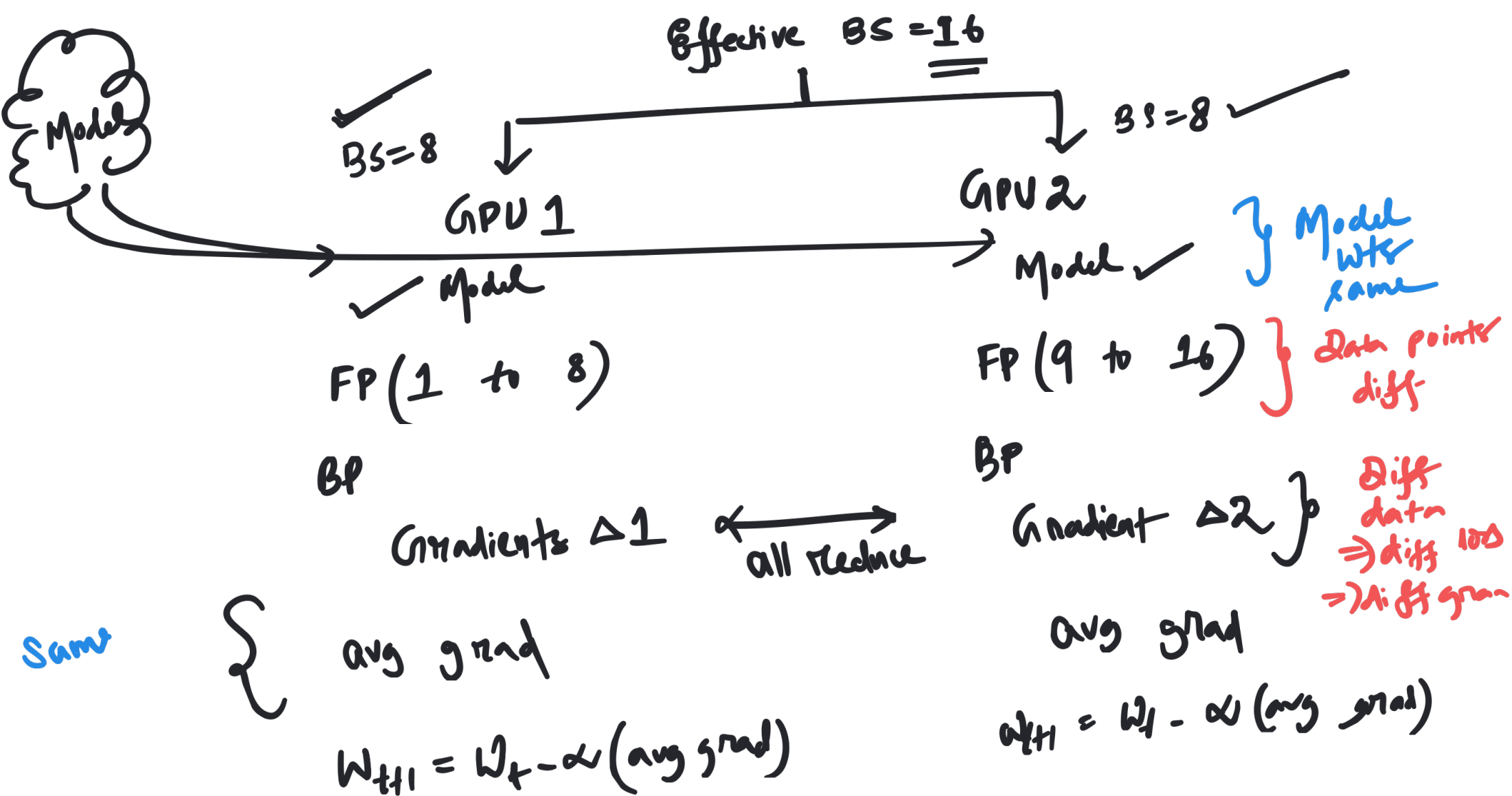
Model + SBS (10) → GPU → No OOM

Batch

B1 $m(FP + BP)$ ✓ wt

B2 $m(FP + BP)$ wt

} parallelite



Advantage (DDP)

Lower Effective Batch Size

7 day (1000)

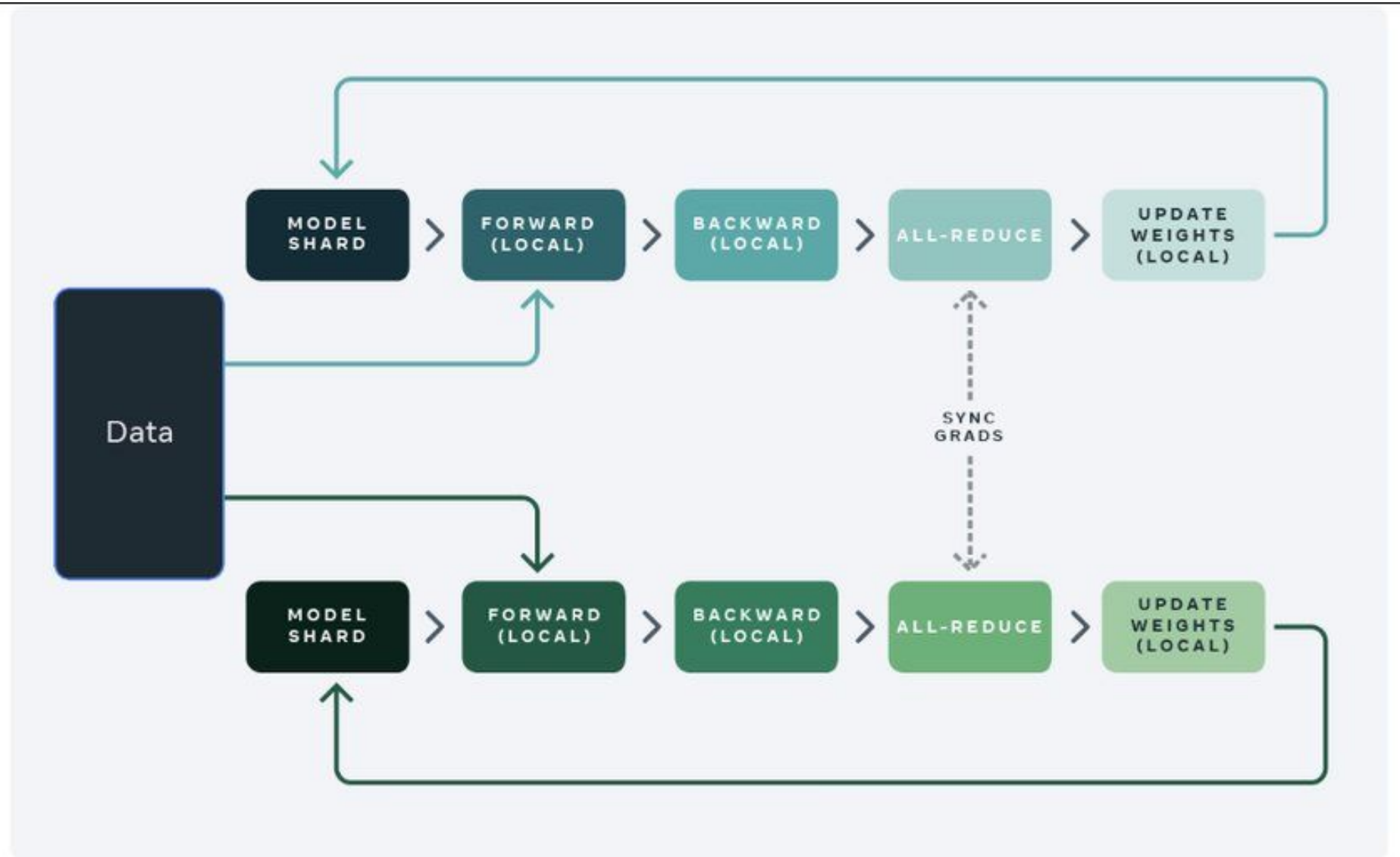
7 GPUs (1 day)

7 day (7000)

7 GPUs (1000) ^{each} 1 day

Disadvantage (DDP)

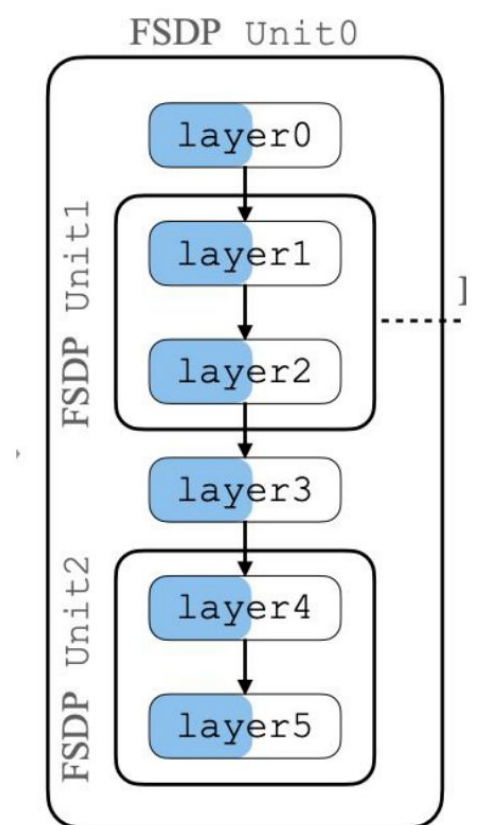
- large model to fit the entire GPU
- not distributing model wts



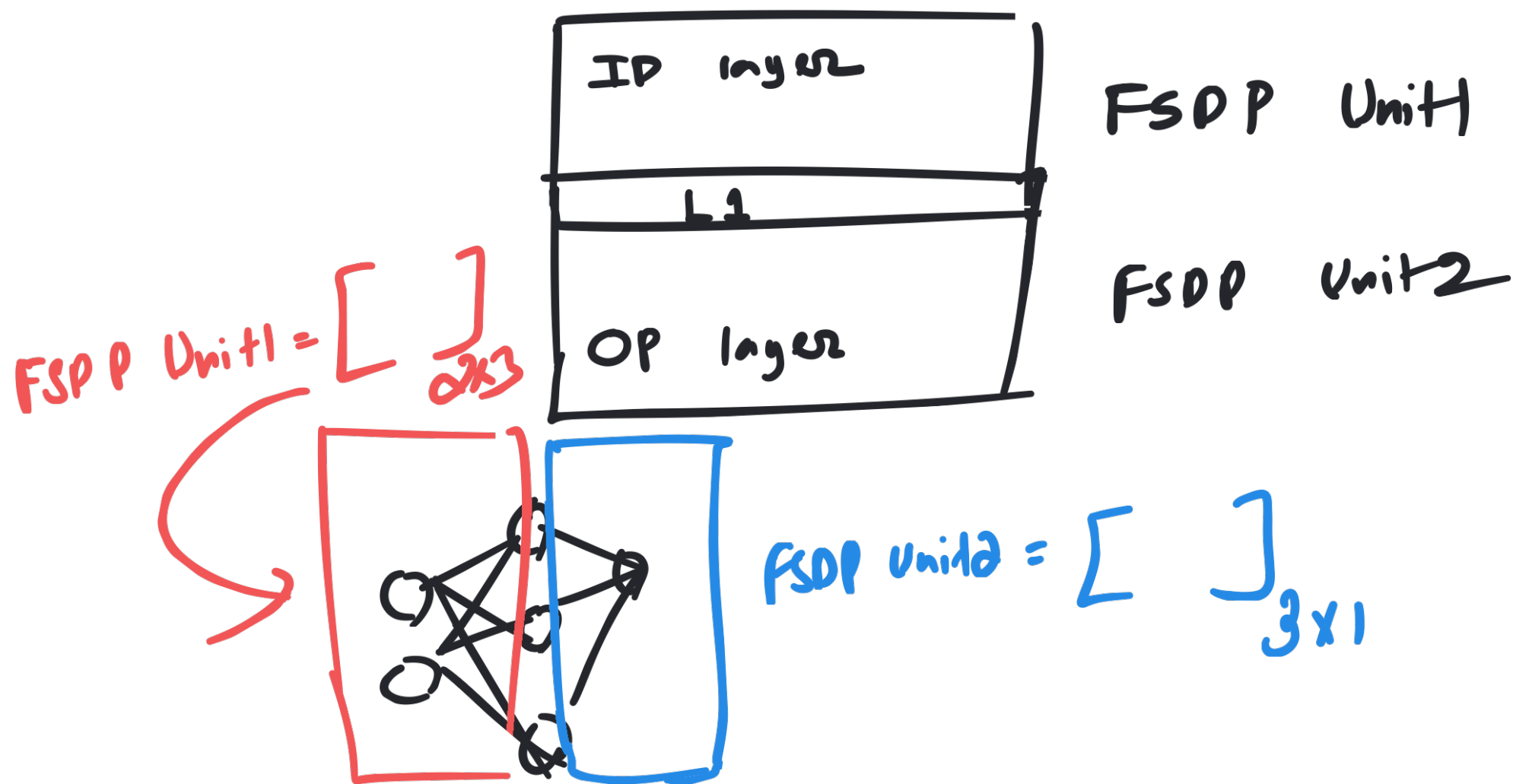
✓ Fully Sharded Data Parallel

FSDP Unit [Vertical “Splitting”]

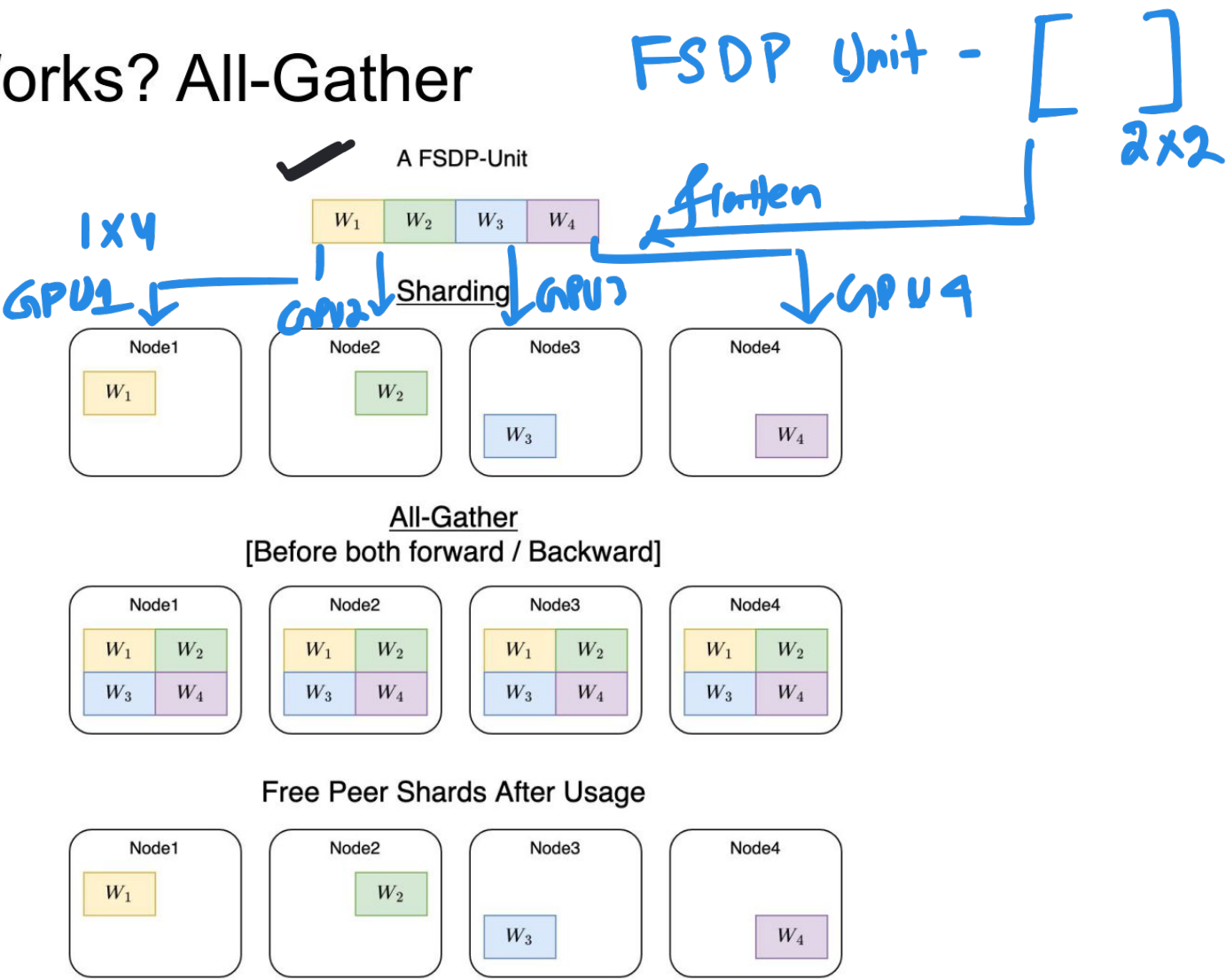
- A unit to split the model. E.g.,
 - A layer
 - A stage
 - A group of layers (nn.Module)



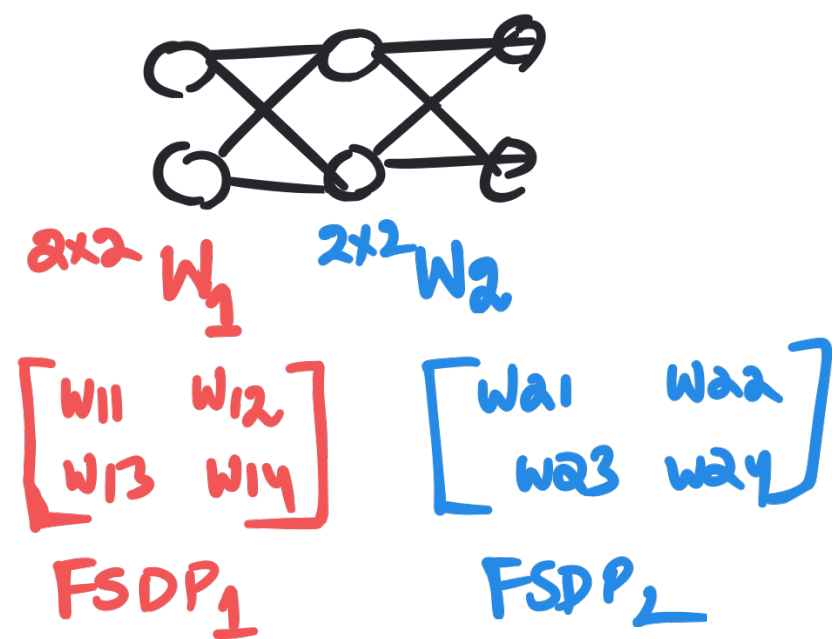
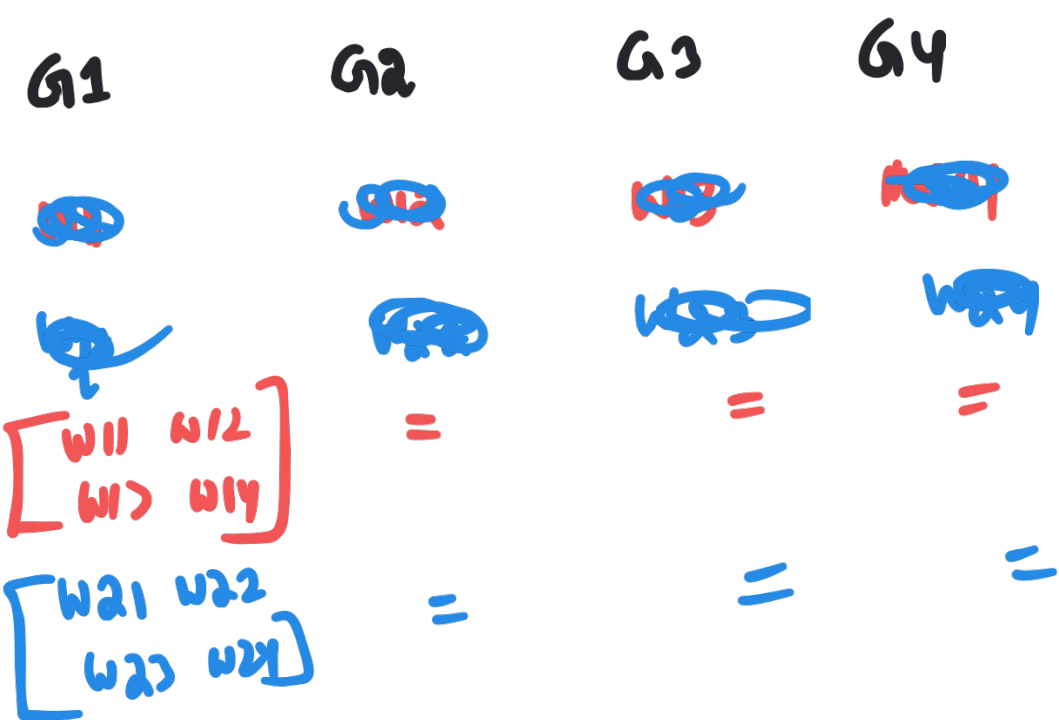
Step 1 Model $\rightarrow$ Divide into FSDP Units



How FSDP Works? All-Gather



Distributed Data Parallel



10 Layers

1 layer
complete

9 layers
shuffled

DDP

$$G1 \begin{bmatrix} 11 & 12 \\ 13 & 14 \end{bmatrix} \begin{bmatrix} 21 & 22 \\ 23 & 24 \end{bmatrix}$$

$$G2 \begin{bmatrix} 11 & 12 \\ 13 & 14 \end{bmatrix} \begin{bmatrix} 21 & 22 \\ 23 & 24 \end{bmatrix}$$

Model

$$\begin{bmatrix} 11 & 12 \\ 13 & 14 \end{bmatrix} 41 \begin{bmatrix} 21 & 22 \\ 23 & 24 \end{bmatrix} 12$$

FSDP

$$G1 \begin{bmatrix} 11 \\ 13 \\ 21 \\ 23 \end{bmatrix} G2 \begin{bmatrix} 12 \\ 14 \\ 22 \\ 24 \end{bmatrix}$$

All 9m

$$\begin{bmatrix} 11 & 12 \\ 13 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 11 & 12 \\ 13 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 21 \\ 23 \end{bmatrix}$$

$$\begin{bmatrix} 22 \\ 24 \end{bmatrix}$$

BS=40

$\textcircled{1-10}$ G_1 $\textcircled{11-20}$ G_2 $\textcircled{21-30}$ G_3 $\textcircled{31-40}$ G_4

w_{11}

w_{12}

w_{13}

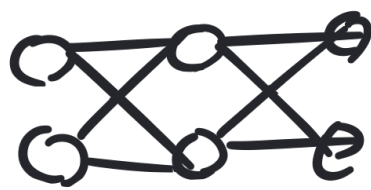
w_{14}

w_{21}

w_{22}

w_{23}

w_{24}



2×2 w_1

2×2 w_2

All gather
FSDP₁

$\begin{bmatrix} w_{11} & w_{12} \\ w_{13} & w_{14} \end{bmatrix}$

$\begin{bmatrix} w_{11} & w_{12} \\ w_{13} & w_{14} \end{bmatrix}$

$\begin{bmatrix} w_{11} & w_{12} \\ w_{13} & w_{14} \end{bmatrix}$

$\begin{bmatrix} w_{11} & w_{12} \\ w_{13} & w_{14} \end{bmatrix}$

$\begin{bmatrix} w_{11} & w_{12} \\ w_{13} & w_{14} \end{bmatrix}$

FSDP₁

$\begin{bmatrix} w_{21} & w_{22} \\ w_{23} & w_{24} \end{bmatrix}$

FSDP₂

Scattered
Reduce

w_{21}

w_{22}

w_{23}

w_{24}

w_{11}

w_{21}

w_{12}

w_{22}

w_{13}

w_{23}

w_{14}

w_{24}

w_{11}

w_{12}

w_{13}

w_{14}

All gather
FSDP₂

$\begin{bmatrix} w_{21} & w_{22} \\ w_{23} & w_{24} \end{bmatrix}$

$\begin{bmatrix} w_{21} & w_{22} \\ w_{23} & w_{24} \end{bmatrix}$

$\begin{bmatrix} w_{21} & w_{22} \\ w_{23} & w_{24} \end{bmatrix}$

$\begin{bmatrix} w_{21} & w_{22} \\ w_{23} & w_{24} \end{bmatrix}$

G_1

w_{11}

w_{21}

G_2

w_{12}

w_{22}

G_3

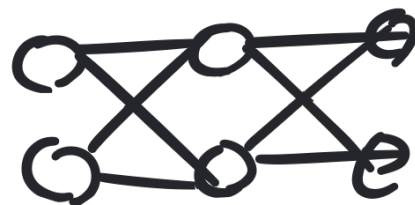
w_{13}

w_{23}

G_4

w_{14}

w_{24}



2×2

w_1

2×2

w_2

$$\begin{bmatrix} w_{11} & w_{12} \\ w_{13} & w_{14} \end{bmatrix}$$

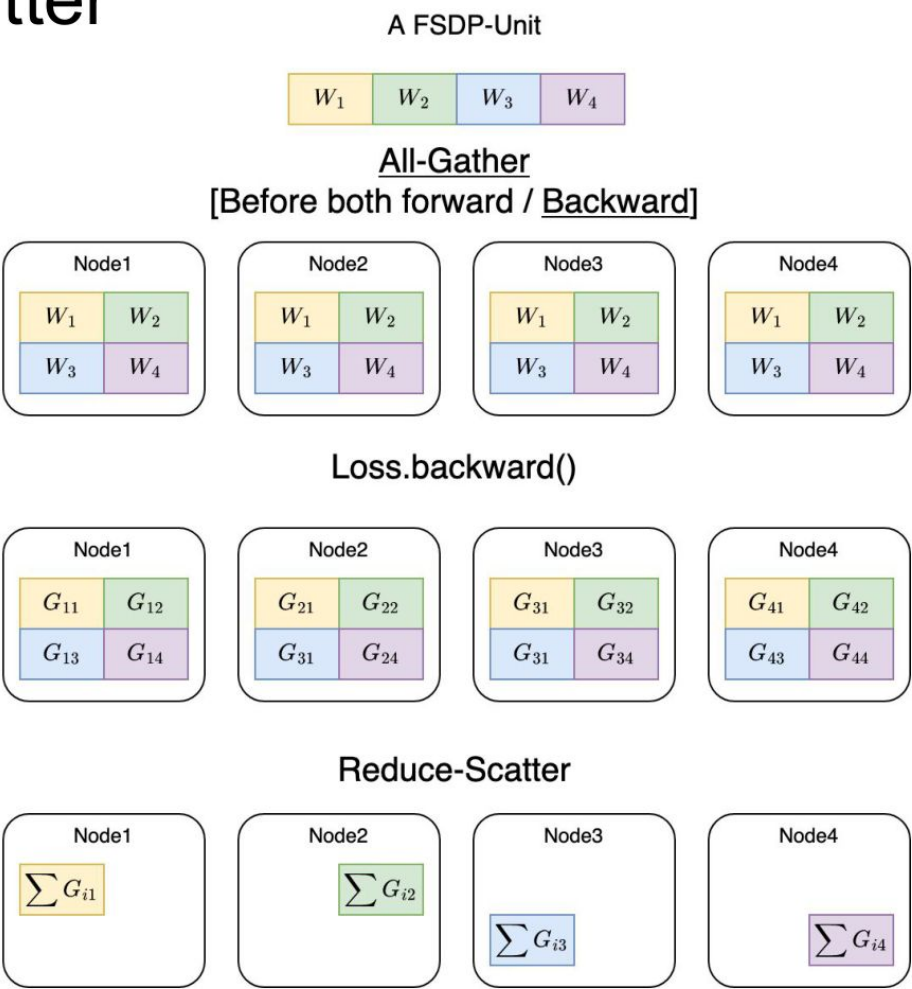
$FSDP_1$

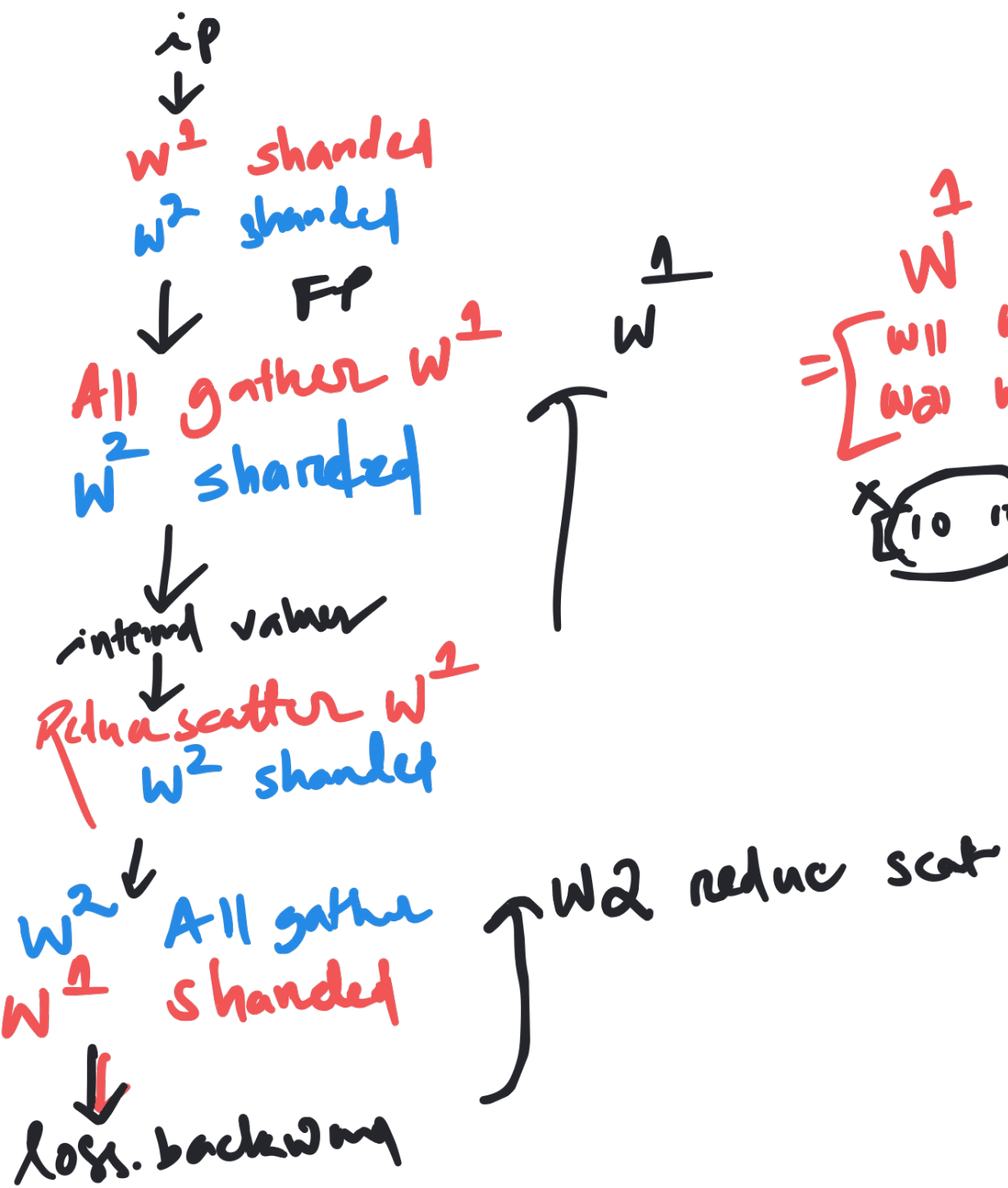
$$\begin{bmatrix} w_{21} & w_{22} \\ w_{23} & w_{24} \end{bmatrix}$$

$FSDP_2$

How FSDP Works? Reduce-Scatter

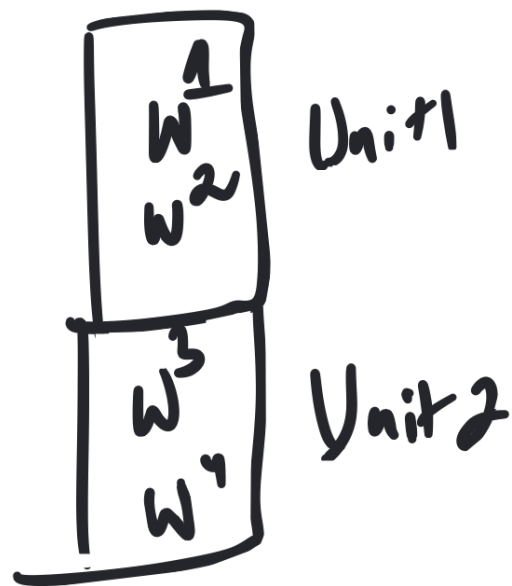
- After all-gather
 - all nodes have the same W_i
 - But different nodes have different G_i
 - Why?

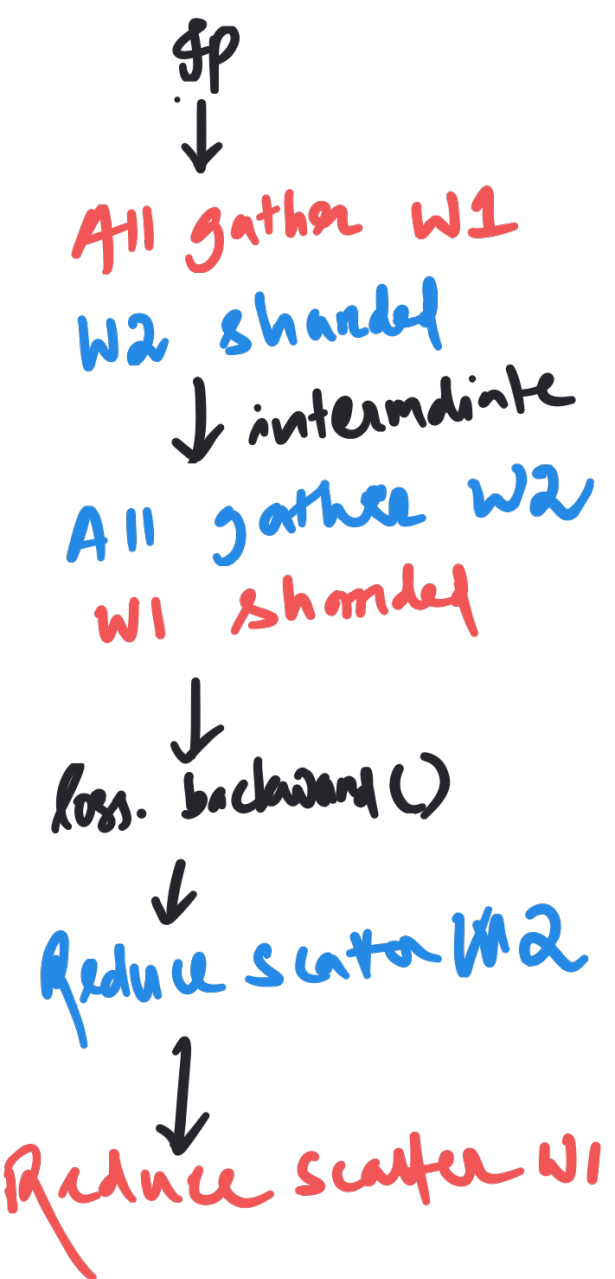




$$= \begin{bmatrix} W_{11} & W_{12} \\ W_{21} & W_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 3 \\ 1 \cdot 2 + 1 \cdot 4 \end{bmatrix}$$





[]_{w1}

[]_{w2}